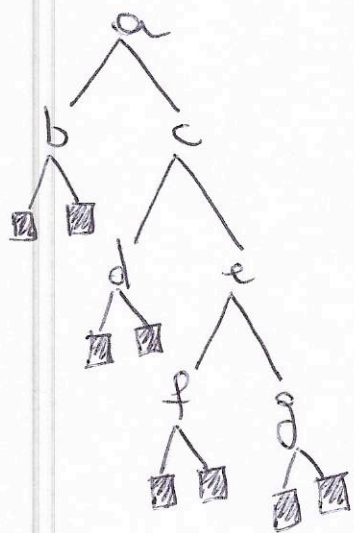


Ex 1



LOADS
→

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	1	0	0	1	1	0	0	1	1	0	0	0	0
1	2	3			4	5			6	7				

"a" → root → $x = 1$ is the "rank" of the visited node

"c" is the right child of "a" → $\text{rank}(c) = \text{rank}_1(2x+1) =$

$$\text{rank}_1(3) = 3$$

↑ $\text{LOADS}[3] = 1 \neq 0$
↳ node exists.

"d" is the left child of "c" → $\text{rank}(d) = \text{rank}_1(2 \cdot 3)$

$$= \text{rank}_1(6) = 4$$

↑ $\text{LOADS}[6] = 1 \neq 0$
↳ node exists

Check ($T_{\text{LOADS}}, L_{\text{LOADS}}, X, d$)

$n = \# \text{nodes in } T;$

for $i = 1$ To n do

if ($L_{\text{LOADS}}[i] == X$) then

$y = i$; $\text{dist} = 0$;

while ($y > 1$) do

$y = \lfloor \text{select}_1(y)/2 \rfloor$;

$\text{dist}++$;

if ($\text{dist} == d$) then return TRUE

return FALSE;

Ex 2

Symbol	0	1	2	3
1	//	//	//	//
2	A	B	//	//
3	C	D	E	//
4	F	G	//	//

FC	
1	2 ← dummy value
2	2
3	1
4	0

$v=0; l=1 \rightarrow v < FC[1] \rightarrow \text{continue}$
 $v=(00)_2=0; l=2 \rightarrow v < FC[2] \rightarrow \text{continue}$
 $v=(000)_2=0; l=3 \rightarrow v < FC[3] \rightarrow \text{continue}$
 $v=(0001)_2=1; l=4 \rightarrow v > FC[4] \rightarrow \text{symbol}[l, v-FC[l]] = \text{symbol}[4, 1-0] = G$

$v=1; l=1 \rightarrow v < FC[1] \rightarrow \text{continue}$
 $v=(11)_2=3; l=2 \rightarrow v > FC[2] \rightarrow \text{symbol}[2, 3-2] = B$

Ex 3

S = CABABCA \$

\$ CABABC	A
A \$	C
A B A	C
A B C	B
B A B	A
B C A	A
C A \$	B
C A B	\$

L

MTF $\rightarrow$ A, B, C
 0 $\rightarrow$ C, A, B
 2 $\rightarrow$ C, A, B
 0 $\rightarrow$ B, C, A
 2 $\rightarrow$ A, B, C
 2 $\rightarrow$ A, B, C
 0 $\rightarrow$ A, B, C
 1 $\rightarrow$ B, A, C

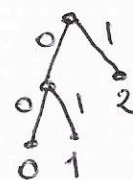
position of \$ (counting from 1)
 $\langle 8, ACCB A A B \rangle$

$\Rightarrow$ RLE ϕ does not modify MTF

To apply Huffman:

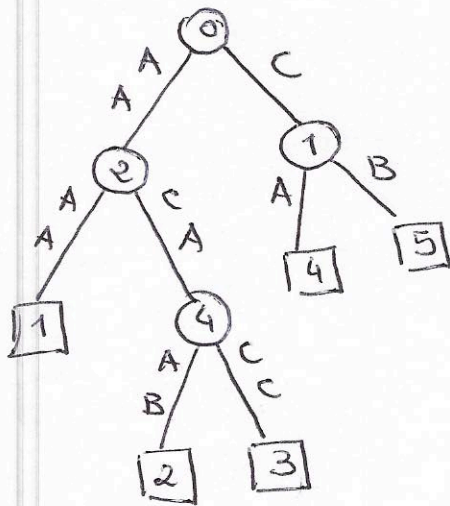
$$p(0) = \frac{3}{7} \quad p(1) = \frac{1}{7} \quad p(2) = \frac{3}{7}$$

$\rightarrow$

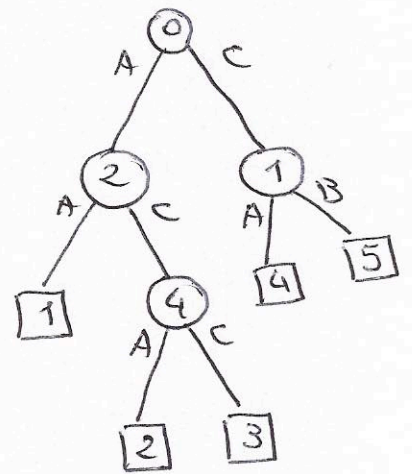


So compressing MTF we obtain: 00100110001

Ex 4

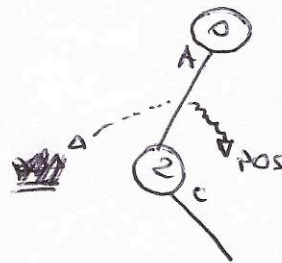


Compacted Trie



Patricia Trie

P1 = ABC



- we thus select a descending leaf, e.g. $\boxed{2}$
AACAAB

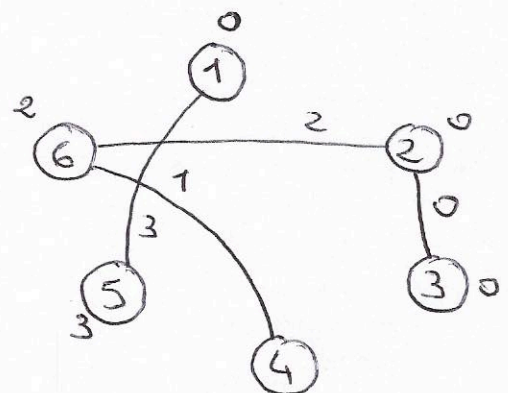
- compute LCP = 1
mismatch P1 $\rightarrow$ B
S₂ $\rightarrow$ A

- so the position of P1 is
To the right of $\boxed{3}$

EX 5

	h	h_1	h_2
11	0	2	3
22	1	4	6
33	2	6	2
44	3	1	5

- * I've not shown the node 0 because this value is not used



It is not acyclic, so the solution for $g()$ does exist

	0	1	2	3	4	5	6
g	1	0	0	0	3	3	2

$$h(t) = g(h_1(t)) + g(h_2(t)) \pmod 4$$